

EXPERIMENT 24 : DATA SEGMENTATION BY COBWEB – HIERARCHIAL CLUSTERING ALGORITHM USING WEKA TOOL

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **Cobweb** -A 1.0 -C 0.0028209479177387815 -S 42

Cluster mode: ☒ Use training set

Result list (right-click for options): 192430 - Cobweb

Clusterer output:

```
--- Run information ---
Scheme:      weka.clusterers.Cobweb -A 1.0 -C 0.0028209479177387815 -S 42
Relation:    iris
Instances:   150
Attributes:  5
  sepalength
  sepalwidth
  petallength
  petalwidth
  class
Test mode:   evaluate on training data

=== Clustering model (full training set) ===

Number of Merges: 2
Number of splits: 2
Number of clusters: 4

node 0 [150]
| leaf 1 [50]
node 0 [150]
| leaf 2 [50]
node 0 [150]
| leaf 3 [50]

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances
1    50 ( 33%)
2    50 ( 33%)
3    50 ( 33%)
```

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode: ☒ Use training set

Result list (right-click for options): 192430 - Cobweb, 192509 - HierarchicalClusterer

Clusterer output:

```
=== Run information ===
Scheme:      weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Relation:    iris
Instances:   150
Attributes:  5
  sepalength
  sepalwidth
  petallength
  petalwidth
  class
Test mode:   evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
((((((((((((((((((0.0:0.03254,0.0:0.03254):0.00913,(0.0:0.03254,0.0:0.03254):0.00913):0.00332,((0.0:0.02778,0.0:0.02778):0.00476,0.0:0.03254):0.01244):0.0:0.0

Cluster 1
((((((((((((((((((1.0:0.07344,(((1.0:0.06508,1.0:0.06508):0.00066,(1.0:0.05008,1.0:0.05008):0.01566):0.00224,1.0:0.06798):0.00546):0.00188,(1.0:0.07137,(1.0:0.055

Time taken to build model (full training data) : 0.04 seconds

=== Model and evaluation on training set ===

Clustered Instances
0    50 ( 33%)
1   100 ( 67%)
```

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode: ☒ Use training set

Result list (right-click for options): 192430 - Cobweb, 192509 - HierarchicalClusterer

Weka Clusterer Visualizer: 192509 - HierarchicalClusterer (iris)

X: petallength (Num) Y: sepalength (Num)

Colour: Cluster (Nom) Select Instance

Plot: iris_clustered

Class colour: cluster0 cluster1

Status: OK